

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q13-0203

Stafford 138 kV
January 16, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0203. The Interconnection Customer proposes a 108.2 MW solar photovoltaic facility interconnecting at the Stafford 138 kV substation in the Pennsylvania area. The request was received on February 04, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 158 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$19,431,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

AMERICAN – AES WARRIOR RUN COGENERATION FACILITY 138 kV: 170.1% loading (430 MVA) following the outage of the AMERICAN – RIDGELEY 345 kV; DFAX = 5.2%.

AMERICAN 345/138 kV TX: 167.0% loading (437 MVA) following the outage of the RIDGELEY – AMERICAN 345 kV; DFAX = 5.2%.

AES WARRIOR RUN COGENERATION FACILITY – RIDGELEY 138 kV: 148.4% loading (620 MVA) following the outage of the AMERICAN – RIDGELEY 345 kV line; DFAX = 5.8%.

For the loss of CHAMBERSBURG – FAIR OAKS 345 kV, the CARREL 345-138 kV TX loads to 133.3% of its 490 MVA rating. Project DFAX: 5.6%.

The SPINDLE SUBSTATION 500/345kV Transformer Bank is overloaded to 122.9% of its 750 MVA emergency rating for the loss of the GREENLAND GAP – SANTEE COOPER 345 kV. The project contributes 18.6% DFAX (20.1 MW) to this overload.

CRONET – FORT HUMBUG 500 kV: 109.5% loading (1,936 MVA) following the outage of the LITTLE CHUTE – FORT HUMBUG 345 kV; DFAX = 8.3%.

For the loss of HILLTOP 500/345kV XFMR, the TALENENERGY MARTINS CREEK LLC WILLIAMSPORT 500/345kV TX loads to 106.3% of its 1,383 MVA rating. Project DFAX: 5.6%.

For the loss of GREENLAND GAP – SANTEE COOPER 345 kV, the ALLIGATOR 500/345 kV Transformer Bank loads to 104.9% of its 752 MVA rating. Project DFAX: 30.9%.

LITTLE CHUTE – TALENENERGY MARTINS CREEK LLC HARRISBURG 345 kV: 104.9% loading (1,626 MVA) following the outage of the FORT HUMBUG – CRONET 500 kV; DFAX = 6.3%.

For the loss of FORT HUMBUG – WEST CAMPUS 345 kV, the ORCHARD RD – FT HUMBUG 500 kV loads to 104.4% of its 3,014 MVA rating. Project DFAX: 9.7%.

The WEST CAMPUS – EDES TAP 2 345 kV Line is overloaded to 103.2% of its 1,155 MVA emergency rating for the loss of the ORCHARD RD – SHAWILLE STA 500 kV. The project contributes 6.4% DFAX (6.9 MW) to this overload.

W CAMPUS – FT HUMBUG 345 kV: 103.1% loading (2,193 MVA) following the outage of the FT HUMBUG – ORCHARD RD 500 kV Line; DFAX = 7.4%.

For the loss of TALENENERGY MARTINS CRK LLC WILLIAMSPORT 500 kV/345 kV Autotransformer, the HILTOP 500/345 kV Transformer Bank loads to 102.6% of its 1,385 MVA rating. Project DFAX: 5.3%.

For the loss of SUNRISE CTR – MERRET 500 kV, the NORTH ALLEGHENY WINDPOWER PROJECT – WAHIAWA 345 kV L loads to 102.3% of its 1,461 MVA rating. Project DFAX: 6.6%.

For the loss of CAYUCOS – SUNRISE CTR 500 kV, the SUNRIS CENTER – LUTESVILLE SUBSTATION 345 kV line loads to 102.3% of its 1,202 MVA rating. Project DFAX: 7.4%.

The FRONT ROYAL – URBAN 345 kV is overloaded to 102.1% of its 1,270 MVA emergency rating for the loss of the INTERNATIONAL PAPER VALLIANT OK – URBAN SUBSTATION 345 kV. The project contributes 9.2% DFAX (10.0 MW) to this overload.

LUTESVILLE – EDES TAP 2 345 kV: 101.1% loading (1,191 MVA) following the outage of the ORCHARD ROAD – SHAWVILLE STATION 500 kV; DFAX = 6.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
AES WARRIOR RUN COGENERATION FACILITY – AMERICAN 138 kV Line	Rebuild line with higher-capacity conductor	\$2,969,000	7% (\$197,000)
AMERICAN 345/138kV Auto	Replace transformer with higher-rated unit	\$11,261,000	7% (\$745,000)
AES WARRIOR RUN COGENERATION FACILITY – RIDGELEY 138 kV	Rebuild line with higher-capacity conductor	\$2,989,000	4% (\$121,000)
CARREL 345/138 kV Transformer	Replace transformer with higher-rated unit	\$16,264,000	21% (\$3,464,000)
SPINDLE SUBSTATION 500/345kV Transformer Bank	Replace transformer with higher-rated unit	\$52,836,000	6% (\$3,219,000)
FORT HUMBUG SUBSTATION – CORONET 500 kV line	Reconductor line	\$81,620,000	2% (\$1,799,000)
TALENENERGY MARTINS CREEK LLC WILLIAMSPORT 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$42,791,000	4% (\$1,565,000)
ALLIGATOR 500 kV/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$24,375,000	15% (\$3,592,000)
LITTLE CHUTE – TALENENERGY MARTINS CREEK LLC HARRISBURG 345 kV	Reconductor line	\$8,731,000	2% (\$175,000)
ORCHARD RD – FT HUMBUG 500 kV	Reconductor line	\$20,771,000	2% (\$326,000)
WEST CAMPUS – EDES TAP 2 345 kV Line	Reconductor line	\$14,477,000	2% (\$346,000)
W CAMPUS – FORT HUMBUG SUBSTATION 345 kV Line	Reconductor line	\$35,459,000	2% (\$627,000)
HILLTOP 500 kV/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$20,599,000	4% (\$894,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NORTH ALLEGHENY WINDPOWER PROJECT – WAHIAWA 345 kV L	Reconductor line	\$14,541,000	3% (\$467,000)
SUNRIS CENTER – LUTESVILLE SUBSTATION 345 kV line	Reconductor line	\$9,115,000	3% (\$288,000)
FRONT ROYAL – URBAN 345 kV	Reconductor line	\$38,134,000	4% (\$1,328,000)
LUTESVILLE – EDES TAP 2 345 kV	Reconductor line	\$11,157,000	2% (\$278,000)
	Total allocated to project		\$19,431,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.